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36 (SEM-II) AS 122

2026

MATHEMATICAL SCIENCE-II

Paper : AS-122

Full Marks : 100

Time : Three hours

**The figures in the margin indicate
full marks for the questions.**

Write the answers to **PART-A** and **PART-B**
in **separate books**.

PART-A

1. Answer **any five** questions : $3 \times 5 = 15$

(i) Compute $\left(\frac{1 + i\sqrt{3}}{2} \right)^9$

(ii) Find the square roots of $2i$.

(iii) Where is $f(z) = \frac{z^2 + z}{(z+1)(z-1)}$ discontinuous? Find the limit given by $\lim_{z \rightarrow i} f(z)$. ✓

(iv) Find $f'(z)$, where $f(z) = z^3$. ✓

(v) Find the points where $f(z) = |z|^2$ is differentiable. ✓

(vi) Show that $e^z \neq 2$ for all complex numbers z .

2. Answer **any three** questions : $5 \times 3 = 15$

(i) Show that $f(z) = \bar{z}$ is not differentiable at any complex number (without using Cauchy-Riemann equations).

(ii) Verify whether the function $f(z)$ is entire or not where

$$f(z) = e^{-y} \sin x - ie^{-y} \cos x$$

(here $z = x + iy$) and hence find $f'(z)$.

(iii) Find all possible values of $\log(-3i)$ and $(1+i)^{-1+i}$.

(iv) Prove that $\sin^2 z + \cos^2 z = 1$ for all $z \in \mathbb{C}$.

(v) Find a Laurent Series expansion of the

function $f(z) = \frac{-1}{(z-1)(z-2)}$ valid in

the domain $1 < |z| < 2$.

3. Answer **any two** questions from below :

10×2=20

(i) ✓ (a) Let $f(z) = \frac{1}{z^2}$, $z \neq 0$. Show that

$f(z)$ is analytic for all $z \neq 0$ and hence find $f'(z)$.

(b) Prove that $f(z) = \sin z$ is an entire function and hence find $f'(z)$.

5+5=10

(ii) Let $u(x, y) = 2 + 3x - y + x^2 - y^2 - 4xy$.

Show that $u(x, y)$ is harmonic in some domain and determine the harmonic conjugate $v(x, y)$; hence find the analytic function

$f(z) = u(x, y) + iv(x, y)$ in terms of z only where $z = x + iy$.

(iii) Evaluate the following integrals :

(a) $\int_C \frac{z+2}{z} dz$, where the contour C

is the semicircle

$$z = 2e^{it}, \pi \leq t \leq 2\pi.$$

(b) $\int_C \frac{dz}{z^2 + 4z + 4}$, where the contour C is positively oriented circle $|z| = 1$.

(c) $\int_C \frac{e^{2z}}{(z-1)^4} dz$, where the contour C is the circle given by $|z| = 3$.

$$3+3+4=10$$

(iv) (a) Find the Taylor series representation of the function $f(z) = e^z$ and specify the region of its validity.

(b) Using Cauchy residue theorem, find the value of the integral

$$\int_C e^{1/z^2} dz, \text{ where } C \text{ is the unit circle } |z| = 1.$$

$$5+5=10$$

PART-B

1. ✓ Evaluate $\iint_D x^2 y dx dy$ where

$$D = \{(x, y) : x^2 + y^2 \leq 1, y \geq 0\} . \quad 6$$

OR

Show that the equation

$$(y^4 + 4x^3 y + 3x) dx + (x^4 + 4xy^3 + y + 1) dy = 0$$

is exact and find its solution.

2. ✓ Evaluate $\int_{-\frac{3}{2}}^{\frac{1}{2}} \int_{x^2}^{-x+\frac{3}{4}} dx dy$ directly (call it I).

Then change the order of integration is the above integral and hence evaluate its value (call it J). Show that $I = J$. 6

3. Evaluate $\iint_D (1 - x^2 - y^2)^{1/2} dx dy$ by

changing to polar coordinates where

$$D = \{(x, y) : x^2 + y^2 \leq 1\} . \quad 6$$

OR

Put the linear first order ODE below in standard form and solve it :

$$x \frac{dy}{dx} + 2y = \frac{dy}{dx} + 4. \quad \checkmark$$

4. Evaluate the vector surface integral

$\iint_S (\vec{f} \cdot \hat{n}) dS$ where $\vec{f} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ and S is that part of the surface of the sphere $x^2 + y^2 + z^2 = 9$ which lies in the first octant $(x, y, z \geq 0)$. 6

OR

Solve the Bessels differential equation

$x^2 y'' + xy' + (x^2 - n^2)y = 0$ in a series of increasing powers of x by Frobenius method. 7.

5. Using Gauss's divergence theorem evaluate

$\iiint_S (\vec{f} \cdot \hat{n}) dS$ where $\vec{f} = x^3\hat{i} + y^3\hat{j} + z^3\hat{k}$ and 8.

S is the surface of the sphere

$$x^2 + y^2 + z^2 = 1. \quad \text{6}$$

OR

Verify Stokes Theorem for the vector function $\vec{f} = (x^2 - y^2)\hat{i} + 2xy\hat{j}$ integrated around the rectangle in $z = 0$ and bounded by the lines $x = 0$, $y = 0$, $x = a$, $y = b$ (where a , b are constants).

6. (i) Solve the second order linear ODE with constant coefficients

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = \sin 2x.$$

- (ii) Solve the Cauchy Euler ODE :

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} = 12 \log x.$$

6

7. Find the volume of the solid bounded by the planes $x = 0$, $y = 0$, $z = 0$ and

$$4x + 2y + z = 8.$$

6

8. Verify Green's Theorem is the plane for

$\oint [(x^3y - 2xy^2) dx - 6x^2y^2 dy]$ where C is the closed curve of the region bounded by the lines $x = 0$, $y = 0$ and $x + 2y = 4$.

8

OR

Apply the method of variation of parameters

to solve $\frac{d^2y}{dx^2} - \frac{1}{x^2}y = \frac{2e^x}{x^2}.$

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